

Why Study Auctions?

Auctions are one of the most important applications of Bayesian Games.

Examples include

- Spectrum auctions
- Google Ads auctions
- Procurement contracts
- IPO pricing
- Art and antique auctions
- Electricity markets

Key Question

If an item is worth 10,000 to you,
how much should you bid?

Auctions as Bayesian Games

Each bidder has private information.

- Type : v_i (private valuation)
- Action: b_i (bid)
- Utility depends on
 - winning,
 - payment,
 - valuation.

Thus, $\boxed{\text{Type} \rightarrow \text{Bid}}$ is a Bayesian strategy.

Components of an Auction

An auction consists of

$$(N, \Theta, B, u)$$

where

- N : bidders
- $\Theta = \{v_i\}$: valuations
- $B = \{b_i\}$: bids
- u_i : utility

Nature first assigns valuations.

Each bidder observes only their own valuation.

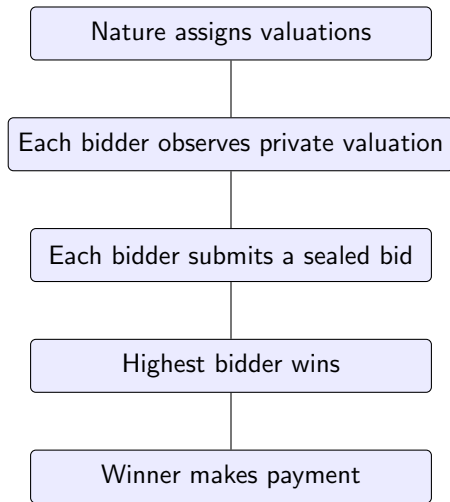
Independent Private Value (IPV) Model

Assumptions

- ① Every bidder has a private valuation v_i .
- ② Valuations are independent.
- ③ Valuations are drawn from $U[0, 1]$.
- ④ Each bidder knows only their own valuation.

This is the most widely studied auction model.

Timeline of an Auction



Bidder's Utility

Suppose bidder i has valuation v_i and submits bid b_i .

Utility is

$$u_i = \begin{cases} v_i - p_i, & \text{if bidder wins,} \\ 0, & \text{otherwise.} \end{cases}$$

where p_i is the payment made.

Major Auction Formats

Auction	Winner Pays
English	Final price
Dutch	Accepted descending price
First-Price Sealed Bid	Own bid
Second-Price (Vickrey)	Second-highest bid

English Auction

Characteristics

- Open ascending bids
- Information revealed gradually
- Highest bid wins
- Winner pays final bid

Examples

- Christie's
- Sotheby's
- eBay (proxy implementation)

Dutch Auction

Characteristics

- Auctioneer starts with a high price.
- Price decreases continuously.
- First bidder to accept wins.

Applications

- Flower markets
- IPO pricing

First-Price Sealed-Bid Auction

Each bidder submits
one sealed bid.

Highest bid wins.

Winner pays
their own bid.

Question

How much should a rational bidder bid?

Second-Price (Vickrey) Auction

Highest bidder wins.

However, the winner pays the Second Highest Bid.

This surprising rule encourages truthful bidding.

Comparison of Auction Formats

	English	Dutch	First	Second
Open	Yes	Yes	No	No
Own Bid Paid	No	Yes	Yes	No
Truthful Bidding	Nearly	No	No	Yes

First-Price Sealed-Bid Auction

Rules

- Every bidder submits exactly one sealed bid.
- Bids are submitted simultaneously.
- The highest bidder wins.
- The winner pays **their own bid**.

Since bidders do not know others' valuations, the auction is naturally modeled as a Bayesian Game.

Bidder's Strategy

Each bidder has

$$v_i \in [0, 1]$$

representing their private valuation.

A strategy is a function

$$b(v) : [0, 1] \rightarrow [0, 1].$$

Instead of choosing one bid,
the bidder chooses a bid for every possible valuation.

Strategy = Bid as a function of valuation.
--

Utility of a Bidder

Suppose bidder i has valuation $v_i =$ and submits bid b_i . Utility is

$$u_i = \begin{cases} v_i - b_i, & \text{if bidder wins,} \\ 0, & \text{otherwise.} \end{cases}$$

Notice the trade-off:

- Bid higher $\Rightarrow$ increase chance of winning.
- Bid higher $\Rightarrow$ lower profit if you win.

Expected Utility

Winning is uncertain.
Therefore,

$$\text{Expected Utility} = \text{Profit} \times \text{Probability of Winning.}$$

Thus,

$$EU(b) = (v - b)P(\text{Win}).$$

Finding the optimal bid requires maximizing expected utility.

Probability of Winning

Assume

- Two bidders.
- Valuations are uniformly distributed on

$[0, 1]$.

Suppose your bid is b . You win whenever the opponent bids less than b
Hence,

$$P(\text{Win}) = P(B_j < b).$$

Uniform Valuations

Suppose bidders use $b(v) = kv$, where $0 < k < 1$. Then $v = \frac{b}{k}$.
Since, $v \sim U(0, 1)$, we obtain:-

$$P(\text{Win}) = \frac{b}{k}.$$

This converts expected utility into a function of the bid.
Substituting,

$$EU(b) = (v - b) \frac{b}{k}.$$

Expected Utility Function

Expanding,

$$EU(b) = \frac{vb - b^2}{k}.$$

The bidder chooses b to maximize this function. Differentiate

$$EU(b) = \frac{vb - b^2}{k}.$$

$$\frac{dEU}{db} = \frac{v - 2b}{k}.$$

Finding the Optimal Bid

Setting

$$\frac{dEU}{db} = 0$$

gives

$$b = \frac{v}{2}.$$

$$\boxed{b(v) = \frac{v}{2}}$$

This is the Bayesian Nash Equilibrium bidding strategy for two bidders.

Independent Private Values (IPV) Model

The **Independent Private Values (IPV)** model is the most widely used framework in auction theory.

Definition

Each bidder has a private valuation of the auctioned object.

$$v_i \in \mathbb{R}_{\geq 0}$$

which is

- known only to bidder i ,
- independent of the valuations of other bidders.

The bidder's valuation represents the maximum amount they are willing to pay.

Assumptions of the IPV Model

The Independent Private Values model relies on the following assumptions.

- ① Each bidder has a private valuation v_i
- ② Valuations are independently distributed.
- ③ Each bidder knows only their own valuation.
- ④ Bidders know the probability distribution of all valuations.
- ⑤ Valuations remain fixed throughout the auction.
- ⑥ Bidders are risk neutral.

Mathematical Formulation

Suppose there are n bidders.

Each bidder's valuation v_i is independently drawn from $F(v)$,
with probability density $f(v)$.

A common assumption is $v_i \sim U[0, 1]$.

Hence,

$$P(v_i \leq x) = x, \quad 0 \leq x \leq 1.$$

Why "Independent" and "Private"?

Private

- Each bidder knows only their own valuation.
- Other bidders cannot observe it.

Independent

- One bidder's valuation does not affect another's.
- Knowing another bidder's valuation provides no information about yours.

Examples include

- Artwork for personal enjoyment
- Collectibles
- Customized machinery

Example

Three bidders have private valuations

$$v_1 = 85, \quad v_2 = 60, \quad v_3 = 95.$$

These valuations are

- privately known,
- independent,
- fixed before bidding begins.

Although bidder 3 values the item the most, they may choose to bid less than 95 depending on the auction format.

Key Question

How should each bidder convert their valuation into a bid?

From Valuation to Bid

Knowing your valuation is not enough.

A rational bidder must determine

Optimal Bid =?

This leads to one of the central questions in auction theory:

Optimal Bidding Strategy

Given a private valuation v , what bid $b(v)$ maximizes the bidder's expected utility?

Objective

Consider a First-Price Sealed-Bid Auction.

Assume

- n bidders
- Independent Private Values
- Risk-neutral bidders
- $v_i \sim U[0, 1]$

Question:

Given valuation v , what bid $b(v)$ maximizes expected utility?

Utility Function

Suppose bidder i has valuation v . If they bid b , their utility is

$$u = \begin{cases} v - b, & \text{if they win,} \\ 0, & \text{otherwise.} \end{cases}$$

Therefore,

$$EU(b) = (v - b) \times P(\text{Win})$$

The only remaining task is to compute

$$P(\text{Win}).$$

Probability of Winning

Assume every bidder uses the same increasing bidding strategy $b(v)$. Suppose bidder i submits bid B .

They win whenever $b > b(v_j)$ for every opponent.

Since

$$b(v)$$

is increasing, this is equivalent to

$$v_j < b^{-1}(B).$$

Probability of Winning

Since , $v_j \sim U[0, 1]$,
we have

$$P(v_j < x) = x.$$

Hence,
the probability that one opponent bids below you is $b^{-1}(B)$.
There are $n - 1$ independent opponents. Therefore,

$$P(\text{Win}) = (b^{-1}(B))^{n-1} . = (b^{-1}(B))^{n-1} .$$

Expected Utility

Substituting, $EU(b) = (v - b) (b^{-1}(B))^{n-1}$.

This expression is difficult to optimize directly. Instead, suppose every bidder follows the same equilibrium strategy

$$b(v).$$

Then

$$B = b(v).$$

Hence,

$$b^{-1}(B) = v.$$

Expected Utility in Equilibrium

The equilibrium utility becomes

$$U(v) = (v - b(v))v^{n-1}.$$

Differentiate with respect to v .

Applying the product rule,

$$U'(v) = v^{n-1} + (n-1)(v - b(v))v^{n-2}.$$

This is the envelope condition.

Differential Equation

Rearranging,
the equilibrium bidding function satisfies

$$b'(v) + \frac{n-1}{v}b(v) = n-1.$$

This is a first-order linear differential equation.

General Equilibrium

For n risk-neutral bidders with independent private valuations

$$U(0, 1),$$

the symmetric equilibrium bidding strategy is

$$b(v) = \frac{n-1}{n} v.$$

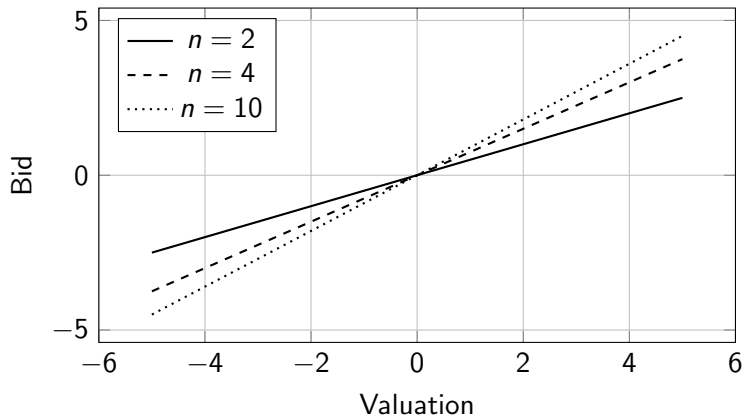
Examples

$$n = 2 \Rightarrow b(v) = \frac{1}{2} v.$$

$$n = 5 \Rightarrow b(v) = \frac{4}{5} v.$$

As competition increases, bidders bid closer to their true valuations.

Equilibrium Bidding Functions



Python Simulation

```
import numpy as np

v = np.linspace(0,1,11)

for n in [2,3,5,10]:

    bids = ((n-1)/n)*v

    print(f"\n{n} bidders")

    print(bids)
```

Observe that equilibrium bids increase as the number of bidders increases.

Second-Price (Vickrey) Auction

Rules

- Every bidder submits one sealed bid.
- The highest bidder wins.
- The winner pays the **second-highest bid**.

Question

Should you still bid below your valuation?

Utility in a Second-Price Auction

Suppose v_i = valuation, b_i = bid.

If bidder i wins, the payment equals the second-highest bid $b_{(2)}$.

Hence

$$u_i = \begin{cases} v_i - b_{(2)}, & \text{if bidder wins,} \\ 0, & \text{otherwise.} \end{cases}$$

Notice that

Your payment does not depend on your own bid.

Why Truthful Bidding?

Suppose your valuation is

$$v = 80.$$

You have three possible choices

$$b < 80, \quad b = 80, \quad b > 80.$$

Which one maximizes your utility?

We analyze all possible cases.

Case Analysis

Assume your valuation is $v = 80$. Opponent's highest bid is 70. If you bid 80, you win and pay 70.

$$Utility = 80 - 70 = 10.$$

If you bid 90, nothing changes. Utility remains 10. Bidding higher gives no additional benefit.

Why Overbidding Can Be Dangerous

Suppose $v=80$, Opponent bids 90.

Truthful bidding loses the auction.

Utility = 0

Now suppose you bid 100.

You win, pay 90, and obtain utility of :-

$$80 - 90 = -10.$$

Overbidding can lead to negative utility.

Why Underbidding Can Be Harmful

Suppose $v=80$, Opponent bids 75.

Truthful bid wins.

Utility

$$80 - 75 = 5.$$

If you underbid 70, you lose. Utility is 0.

Thus, Underbidding may lose profitable opportunities.

Dominant Strategy

Combining all possible cases,

Strategy	Outcome	Utility
Bid Truthfully	Never Worse	Optimal
Overbid	Risk of Overpaying	Lower
Underbid	Lose Valuable Items	Lower

$$b(v) = v$$

is a dominant strategy.

Vickrey's Theorem

Theorem

In a second-price sealed-bid auction, truthful bidding is a dominant strategy.

Consequently, every bidder should submit their true valuation.

$$b(v) = v.$$

The strategy is independent of other bidders' actions.

First-Price vs Second-Price Auctions

Property	First Price	Second Price
Winner Pays	Own Bid	Second Bid
Truthful Bidding	No	Yes
Optimal Bid	Shade Bid	True Value
Dominant Strategy	No	Yes
Bayesian Nash Equilibrium	Yes	Truthful

Worked Example

Three bidders have valuations

100, 80, 60.

First Price

Winner pays 100.

Utility = 0

Second Price

Winner pays 80.

Utility = $100 - 80 = 20$.

The auction rewards truthful bidding.

Python Simulation

```
bids = [100,80,60]

winner = max(bids)

payment = sorted(bids)[-2]

utility = winner-payment

print("Winner:",winner)

print("Payment:",payment)

print("Utility:",utility)
```

Modify the bids and compare with the first-price auction.

An Interesting Question

We have studied two auction formats.

- First-Price Auction
- Second-Price Auction

Both appear to produce very different bidding strategies.

$$b(v) = \frac{n-1}{n}v$$

versus

$$b(v) = v.$$

Question

Do these auction formats generate the same expected revenue for the seller?

Do Different Auctions Generate the Same Revenue?

Consider the following auction formats:

- First-Price Sealed-Bid
- Second-Price (Vickrey)
- English Auction
- Dutch Auction

Although bidders behave differently in each format, does the seller earn the same expected revenue?

Central Question

Can different auction mechanisms produce the same expected seller revenue?

Revenue Equivalence Theorem

Theorem (Revenue Equivalence)

Suppose

- *bidders are risk neutral,*
- *valuations are independently and identically distributed,*
- *valuations are private,*
- *the bidder with the highest valuation always wins.*

Then,

every auction mechanism satisfying these assumptions produces the same expected revenue.

Assumptions Behind the Theorem

Revenue Equivalence relies on several strong assumptions.

- ① Independent Private Values (IPV)
- ② Risk-neutral bidders
- ③ Symmetric bidders
- ④ Highest valuation always wins
- ⑤ Zero reserve price
- ⑥ No collusion

If these assumptions fail,
Revenue Equivalence may no longer hold.

Intuition Behind Revenue Equivalence

Different auction formats change

- how bidders bid,
- how much winners pay.

However,

strategic bid shading in a First-Price Auction
exactly offsets
the truthful bidding in a Second-Price Auction.

Therefore,

the seller receives the same expected revenue.

Example

Suppose two bidders have valuations : 100,80.

First Price

Equilibrium bid : $b(v) = \frac{1}{2}v$.

Hence 50,40.

Seller revenue : 50.

Second Price

Truthful bids: 100,80.

Seller receives 80.

Individual realizations differ.

Revenue Equivalence concerns the **expected revenue over many auctions**.

Python Simulation

```
import random
N = 100000
fp = 0
sp = 0
for _ in range(N):
    v1 = random.random()
    v2 = random.random()
    fp += max(v1,v2)/2
    sp += min(v1,v2)
print(fp/N)
print(sp/N)
```

Both averages converge to approximately

$$\frac{1}{3}.$$

When Does Revenue Equivalence Fail?

The theorem may fail if

- bidders are risk averse,
- valuations are correlated,
- values are common rather than private,
- reserve prices are imposed,
- bidders collude,
- bidders are asymmetric.

Many real-world auctions violate one or more of these assumptions.

Private Values vs Common Values

Private Value	Common Value
Each bidder has their own valuation	All bidders estimate the same underlying value
Value differs across bidders	True value is identical for everyone
Example: Art for personal enjoyment	Example: Oil field auction

Common-value auctions introduce a new phenomenon: the **Winner's Curse**.

Winner's Curse

Suppose the true value of an oil field is unknown.
Each company receives an estimate.

98, 101, 103, 110, 95.

The company bidding 110 wins.

Why?

Because it was the most optimistic estimate.

Winning the auction may actually indicate that you have overestimated the item's value.

This is the Winner's Curse.

Python: Winner's Curse Simulation

```
import random

true_value = 100

estimates = [
    true_value + random.gauss(0,10)
    for _ in range(5)
]

winner = max(estimates)

print(estimates)

print("Winning Estimate:",winner)

print("Overestimate:",
winner-true_value)
```